VW-MTC1 MazeTrack Controller

Wire-maze motor-coordination assessment controller, ESP32-S3, 4-channel Hall sensing, Li-ion powered

Voltwyn | Product datasheet | Hardware Rev A | Document 20261019-VW-MTC1-Datasheet-Rev00 | 19 October 2026 | Preliminary

Features

- ESP32-S3-WROOM-1 (N8R2): dual-core 240 MHz, Wi-Fi 4 and Bluetooth LE 5, native USB (CDC + JTAG)
- 4 Hall channels, each with ESD protection, handle-absent detection and a 2nd-order Sallen-Key low-pass (232 Hz, Q 0.73)
- External 12-bit SAR ADC (MCP3204), ratiometric to the sensor supply; backup path on ESP32 ADC1
- Switched handle supply (+3V3_H) for low standby drain; I2C bus and haptic motor drive to the handle
- Maze wire / start / finish contact inputs: 9 uA maximum through a person, mains-hum immune, Schmitt-buffered
- 6-axis IMU (ISM330DHCX) for tremor and 3D handle motion
- USB-C power with PTC, TVS and data-line ESD; 1S Li-ion charging at 400 mA (LTC4054)
- Automatic USB/battery power path; low-noise 1 A buck-boost (LTC3536), 91-94 % simulated efficiency
- microSD logging, OLED header, 3 status LEDs, 3 buttons, buzzer, Qwiic expansion
- 90 x 60 mm, 4-layer, single-sided SMT assembly, 4x M3 holes

Applications

- Objective upper-limb coordination testing (wire-maze tasks) for ataxia research
- Rehabilitation games with visual, audio and haptic feedback
- Longitudinal home or clinic logging via microSD, USB or BLE
- University teaching and research platforms

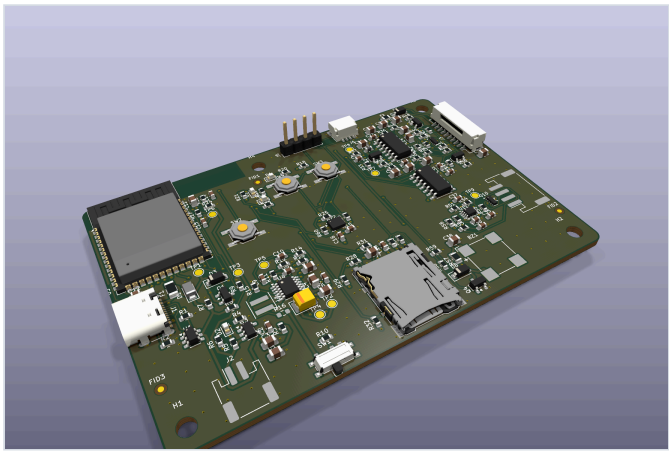
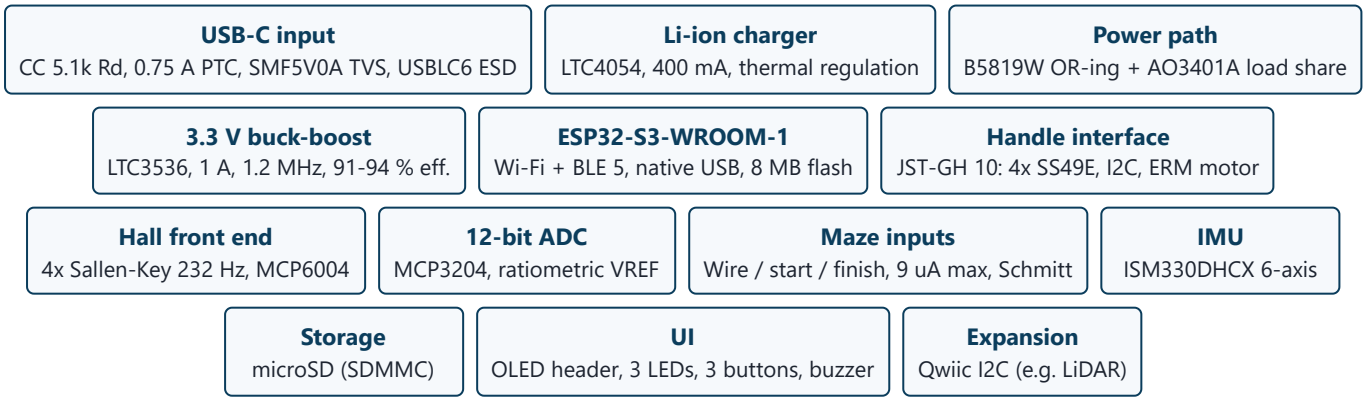


Figure 1. VW-MTC1 Rev A, 3D render from the KiCad layout

Functional block diagram



Regulatory status. Research and education use only. VW-MTC1 is not a registered medical device and must not be used for diagnosis or treatment decisions until assessed under the TGA framework and IEC 60601-1 / ISO 14971. Specifications are design targets from calculation and LTspice simulation; Rev A has not yet been built or tested.

Recommended operating conditions

Parameter	Min	Typ	Max	Unit	Note
USB VBUS input	4.5	5.0	5.5	V	USB-C sink, default USB current (5.1k Rd)
Battery voltage (1S Li-ion)	3.0	3.7	4.2	V	Cell must include a protection PCM
Battery charge current	-	400	-	mA	R_PROG 2.49k; 407.9 mA simulated
Charge float voltage	-	4.2	-	V	4.217 V simulated
Operating ambient (charging)	0	25	40	C	Charger thermal regulation may reduce current above ~45 C
Operating ambient (discharging)	0	25	50	C	Design target
Handle cable length	-	1.0	1.5	m	I2C at 100-400 kHz with 2.2k pull-ups

Electrical characteristics

Values marked SIM are LTspice results (see the design report and `Simulation/results.json`). CALC values are hand calculations. Rev A hardware has not yet been measured.

Parameter	Condition	Value	Basis
+3V3 output voltage	FB 1M / 221k	3.315 V nominal	CALC
+3V3 load-step excursion	50 to 600 mA step, VIN 3.0-4.6 V	3.197 to 3.396 V	SIM01b
+3V3 ripple	300 mA, PWM, VIN 3.7 V	37 mV p-p	SIM01c
Buck-boost efficiency	VIN 3.7 V; 50 / 300 / 600 mA	93.9 / 93.9 / 91.0 %	SIM01c
Switching frequency	RT tied to VIN	1.2 MHz	Datasheet
+3V3 during USB unplug/replug	300 mA load	3.262 to 3.419 V	SIM03
Hall filter cut-off	Sallen-Key, 10k/10k, 100n/47n	238 Hz (-3 dB), Q 0.729	SIM04a
Hall filter attenuation	at 1 kHz / 1.2 MHz	-25.4 dB / -45.6 dB	SIM04a
Hall filter group delay	at 10 Hz	0.95 ms	SIM04a
Noise reduction vs original design	broadband + switching noise	9.3 dB SNR gain	SIM04b
ADC resolution	MCP3204, VREF = +3V3_H	12 bit, 0.806 mV/LSB	CALC
Field resolution	SS49E ~0.92 mV/G at 3.3 V	~0.87 G/LSB	CALC
Sample rate	4 channels	2 kS/s per channel (frame time 96 us at 1 MHz SPI)	CALC
Maze input current through a person	3.3 V / (360k + 10k)	8.9 uA max	CALC
Maze input detection latency	1 ohm contact	0.34-0.40 ms	SIM05
Haptic motor drive	20 kHz PWM	3.0 V average (76 % at 3.7 V, 55 % at 4.6 V)	SIM06

Connectors

J1 USB-C (GCT USB4105-GF-A)

Pin	Function
VBUS	5 V in, 0.75 A PTC + TVS
D+ / D-	Native USB, USBLC6 ESD
CC1 / CC2	5.1k Rd (sink)
GND / shield	Ground

J2 Battery (JST PH 2-pin SMD)

Pin	Function
1	VBAT + (1S Li-ion/LiPo with protection PCM)
2	GND

J3 Handle (JST GH 10-pin SMD)

Pin	Function
1	+3V3_H (switched sensor supply)
2-5	HALL_IN1-4 (SS49E outputs)
6	GND / ring
7 / 8	SDA_H / SCL_H
9	MOT+ (VSYS via 0.2 A PTC)
10	MOT- (low-side switched)

J4 Maze (JST PH 4-pin SMD)

Pin	Function
1	MAZE_WIRE
2	MAZE_START pad
3	MAZE_FINISH pad
4	GND

J5 OLED (2.54 mm header)

Pin	Function
1	GND
2	+3V3
3	SCL
4	SDA

J6 Qwiic (JST SH 4-pin)

Pin	Function
1	GND
2	+3V3
3	SDA
4	SCL

J7 microSD (Hirose DM3AT)

Pin	Function
CLK/CMD/D0	SDMMC 1-bit, 10k pull-ups on CMD and D0-D3
DET	Card detect to GPIO14

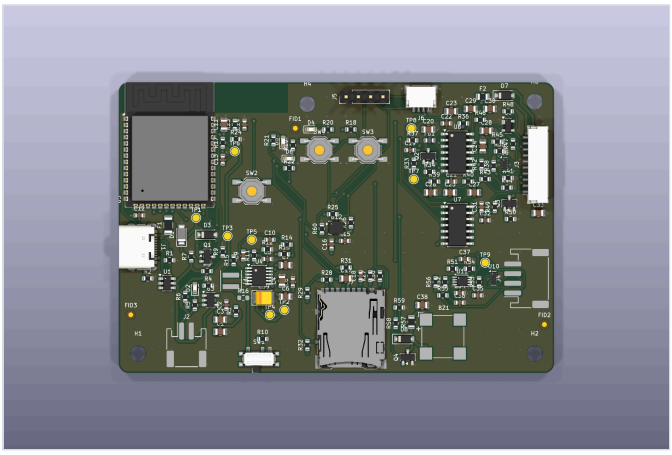
ESP32-S3 GPIO map

GPIO	Signal	Function
GPIO1	VBAT_SENSE	Battery voltage / 2 (ADC1)
GPIO2	VBUS_SENSE	USB present / 2
GPIO3	BTN_START	Start / calibrate button
GPIO0	BOOT / MODE	Boot strap, mode button after boot
GPIO4-7	HALL_F1-F4	Filtered Hall signals, backup ADC1 path
GPIO8 / 9	SDA / SCL	On-board I2C: OLED, IMU, Qwiic
GPIO10-13	ADC CS/MOSI/SCK/MISO	MCP3204 SPI (1 MHz)
GPIO14	SD_CD	Card detect, low = inserted
GPIO15	IMU_INT1	IMU data-ready / wake
GPIO16	HALL_EN_N	Low = handle + analogue rail on

GPIO17 / 18	SDA_H / SCL_H	Handle I2C (2.2k to +3V3_H)
GPIO19 / 20	USB D- / D+	Native USB CDC + JTAG
GPIO21	TOUCH_FINISH	Finish pad contact (low)
GPIO38 / 39 / 40	SD CMD / CLK / D0	microSD, 1-bit SDMMC
GPIO41	TOUCH_WIRE	Ring touched wire (low)
GPIO42	TOUCH_START	Start pad contact (low)
GPIO43	LED_AMB	Idle LED (active high)
GPIO44	CHRG_N	Charging status (low)
GPIO45	BUZZER	Magnetic buzzer drive (~4 kHz PWM)
GPIO46	HAPTIC_PWM	ERM motor 20 kHz PWM
GPIO47	LED_GRN	Forward LED
GPIO48	LED_RED	Reverse LED
GPIO35-37	not connected	Reserved (octal-PSRAM modules)

Mechanical

Item	Value
Board outline	90.0 x 60.0 mm, 2 mm corner radius
Thickness / layers	1.6 mm FR-4, 4 copper layers (signal / GND / +3V3 / signal)
Mounting	4x M3 (3.2 mm) at (4, 56), (86, 56), (86, 4), (39, 4.5) mm from top-left corner
Antenna	ESP32-S3 PCB antenna at the top-left edge; copper-free keepout 35 x 6.3 mm on all layers
Assembly	All components on the top side (single-sided SMT); 1 through-hole header (OLED)
Component count	154 placed footprints, 514 pads, 108 nets



Top view



Bottom view

Revision history

Rev	Date	Change
A / Rev00	19 October 2026	First release of the VW-MTC1 design (not yet built)

Voltwyn VW-MTC1 datasheet. Designed by Sajeevan (Saj) Veeriah. Based on the SEJ446 thesis "Validation for Evaluating Fast Alternating Hand Movements of Ataxia Patients Using a Magnetic Wire Maze Game", Deakin University.